



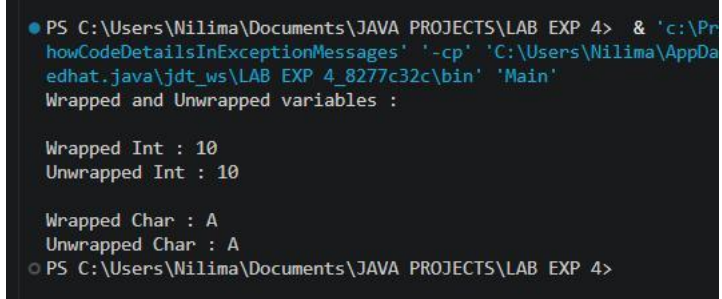
Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 4

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PRN No.	25070122029
Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	11/7/26
Title of Assignment:	Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.
Problem Statement	Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.
Learning Objectives	To perform conversion between primitive data types and wrapper objects using Java wrapper classes.
Theory	Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Program Code	<pre> 1 2 public class Main { 3 public static void main (String[] args) { 4 System.out.println("Wrapped and Unwrapped variables : \n"); 5 6 int x = 10; 7 Integer WrappedX = Integer.valueOf(x); 8 int unwrappedX = WrappedX.intValue(); 9 System.out.println("Wrapped Int : " + WrappedX); 10 System.out.println("Unwrapped Int : " + unwrappedX + "\n"); 11 12 char ch = 'A'; 13 Character ch_wrapped = Character.valueOf(ch); 14 char ch_unwrapped = ch_wrapped.charValue(); 15 16 System.out.println("Wrapped Char : " + ch_wrapped); 17 System.out.println("Unwrapped Char : " + ch_unwrapped); 18 } 19 } </pre>	
Output Screenshots	 <pre> PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 4> & 'c:\Program Files\Java\jdk-8.0.602\bin\java.exe' -cp 'C:\Users\Nilima\AppData\Local\Temp\jdt_ws\LAB EXP 4_8277c32c\bin' 'Main' Wrapped and Unwrapped variables : Wrapped Int : 10 Unwrapped Int : 10 Wrapped Char : A Unwrapped Char : A PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 4> </pre>	
Exercise Qns	Q1. Create a program to convert student marks from String format to Integer and calculate total marks. <pre> 1 // Create a program to convert student marks from String format to Integer and calculate total marks. 2 public class Ex1 { 3 public static void main (String[] args) { 4 String marks1 = "99"; 5 String marks2 = "95"; 6 String marks3 = "87"; 7 8 Float marks_wrapped1 = Float.valueOf(marks1); 9 Float marks_wrapped2 = Float.valueOf(marks2); 10 Float marks_wrapped3 = Float.valueOf(marks3); 11 12 float total_marks = marks_wrapped1 + marks_wrapped2 + marks_wrapped3; 13 float percentage = total_marks / 3; 14 15 System.out.println("Percentage : " + percentage + " %"); 16 System.out.println("Total Marks : " + total_marks + " / 300"); 17 } 18 } </pre> Q2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.	

```

1
2 import java.util.Scanner;
3
4 class Employee {
5     int id;
6     int salary;
7     int bonus;
8
9     Employee (int i, int s, int b) {
10         id = i;
11         salary = s;
12         bonus = b;
13     }
14
15     int net_salary () {
16         return salary + bonus;
17     }
18
19     void display () {
20         System.out.println(x: "\nEmployee Details : ");
21         System.out.println("ID          : " + id);
22         System.out.println("Salary       : Rs." + salary);
23         System.out.println("Bonus        : Rs." + bonus);
24         System.out.println("Net Salary  : Rs." + net_salary());
25     }
26 }

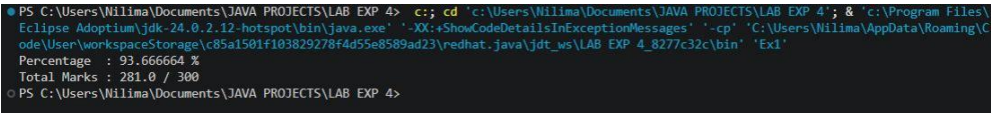
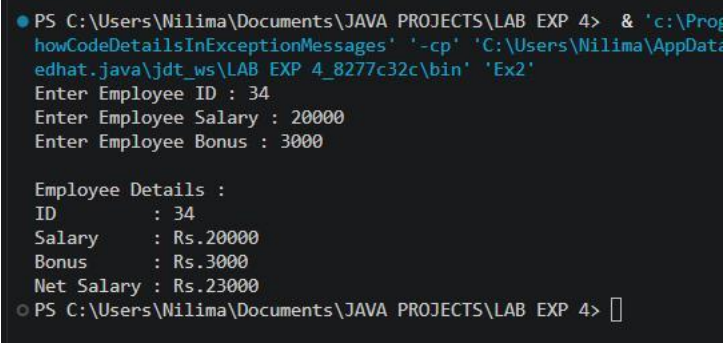
```

Click to add a breakpoint

```

28 public class tx2 {
    Run | Debug
29     public static void main (String[] args) {
30         try (Scanner in = new Scanner(System.in)) {
31             System.out.print(s: "Enter Employee ID : ");
32             String id = in.next();
33             Integer id_wrapped = Integer.valueOf(id);
34
35             if (id_wrapped < 0) {
36                 System.out.println(x: "Employee ID cannot be negative.");
37                 System.exit(status: 0);
38             }
39
40             System.out.print(s: "Enter Employee Salary : ");
41             String sal = in.next();
42             Integer sal_wrapped = Integer.valueOf(sal);
43
44             if (sal_wrapped < 0) {
45                 System.out.println(x: "Employee Salary cannot be negative.");
46                 System.exit(status: 0);
47             }
48
49             System.out.print(s: "Enter Employee Bonus : ");
50             String b = in.next();
51             Integer b_wrapped = Integer.valueOf(b);
52
53             if (b_wrapped < 0) {
54                 System.out.println(x: "Employee Bonus cannot be negative.");
55                 System.exit(status: 0);
56             }
57
58             Employee e = new Employee(id_wrapped, sal_wrapped, b_wrapped);
59             e.display();
60         }
61     }
62 }

```

Output Screenshots	Qn1  Qn 2 
Conclusion	This assignment helped in understanding the concept and importance of Java wrapper classes for converting primitive data types into objects and vice versa. It demonstrated the use of wrapper classes such as Integer, Double, Character, and Boolean for autoboxing, unboxing, parsing, and basic operations on wrapped values. The implementation improved my understanding of data conversion, input validation, and object-oriented programming concepts. These concepts are widely used in Java. Collections Framework, file handling, database applications, and other real-world Java programs where object representation of primitive data is required.